



VNUHCM
University Of Science

REPORT

Introduction to Artificial Intelligence

Lab 2: Logic

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Course section: CSC14003 - 22CLC08

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ASSESSMENT

No.	Details	Self assessment
01	Read input data and store in suitable data structure	100%
02	Implementation of resolution method	100%
03	Inference process and results	100%
04	Test Cases, report, evaluations	100%
Total		100%

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1. Description of functions

1.1. Compile

- After we run the source code, we continue input name of input file and output file as below:

Input file: input1.txt

Output file: output1.txt

PS C:\Users\ThinkPad\Desktop\AI_Group01\Lab02_Logic\21127702> █

- After that, we check the output file in the folder “Output” that we created before:

The screenshot shows a file explorer on the left and a code editor on the right. The file explorer displays a directory structure with folders like `__pycache__`, `Exercise_4`, and `Output`. The `Output` folder is expanded, showing files `output1.txt` through `output6.txt`. The `output1.txt` file is highlighted with a red box. The code editor on the right shows the content of `output1.txt`, which is a list of 11 items:

```
Exercise_4 > Output > output1.txt
You, 37 minutes ago | 1 author (You)
1 3 You, 37 minutes ago • lab2 update
2 -A
3 B
4 -C
5 4
6 -B OR C
7 A OR C
8 A OR -B
9 {}
10 YES
11
```

1.2. Functions

- **negate(w)**: Implement for changing the sign of a proposition, such as '-A' to A or A to '-A'.
- **negateQuery(query)**: Implement for changing the sign of the Query:

```

≡ example.txt
1  # Before
2  C OR B
3
4  # After
5  -C
6  -B
7  

```

- **generateKB(KBClauses, queryClauses)**: Implement for generating the KBclauses and Query, the set of clauses in the CNF representation of $KB \wedge \neg Q$, and change to list. Use negate(w) for changing Query to negative of it.

```

≡ example.txt
1  # Before
2  ## Query
3  -C
4
5  ## KB
6  A OR -B
7  -A OR B OR C
8  B OR C
9  -A OR -C
10
11 # After
12 [['A', '-B'],
13  ['-A', 'B', 'C'],
14  ['B', 'C'],
15  ['-A', '-C'],
16  ['C']]
17

```

```

≡ example.txt
11 # After
12 [['A', '-B'],
13  ['-A', 'B', 'C'],
14  ['B', 'C'],
15  ['-A', '-C'],
16  ['C']]
17
18
19 

```

- **isNegate(a1, a2)**: check for if this proposition is negative to the other.

- **isEquivalent(*clause*)**: check for equivalent clauses or useless clauses such as $(A \vee B \vee \neg B)$.
- **resolventInClauses(*resolvents*, *clauses*)**: check for if resolvent in clauses include its permutations.
- **pl_resolve(*Ci*, *Cj*)**: implement for resolving clauses and return a new resolvent, check if *Ci* has a proposition that negate to a proposition in *Cj*, create a temp of *Ci* and *Cj* and remove this proposition to two of them, and check if this pair temp is not an equivalent clauses or useless clause, return True and new resolvent, just a single proposition in each *Ci*, *Cj*, and these pair is negate to each other, it mean that the resolvent contain an empty clauses and return True and [{"{}"}], else return False and no clause.
- **pl_resolution(*KB*, *query*)**: implement for PL-Resolution, first use the generateKB to generate the set of clauses in the CNF representation of $KB \wedge \neg Q$, we have a loop do finding a resolvent for each pair of this set of clauses, and if pl-resolve found a new clause it means that return True, and use resolventInClauses(*resolvents*, *clauses*) to check the resolvent is in newResolvents (keep resolvents of each loop and empty for new loop) or clauses, if it not in these, we will append the newResolvents in the result list (stand for saving the result of each loop), if the newResolvents has the [{"{}"}] resolvent, it means that we found the answer and return True, if the newResolvents doesn't contain any resolvent ([{"{}"}] also means a resolvent) it means that no new sentence found, the function will return False (KB does not entail α since no new clause is created and no empty clause is found).

REFERENCES

[1] cs221 - rohitapte - Github.com

[cs221/logic/submission.py at master](#)

[2] AI-Lab-PL-Resolution - phungvhbui - Github.com

[phungvhbui/AI-Lab-PL-Resolution - HCMUS](#)